

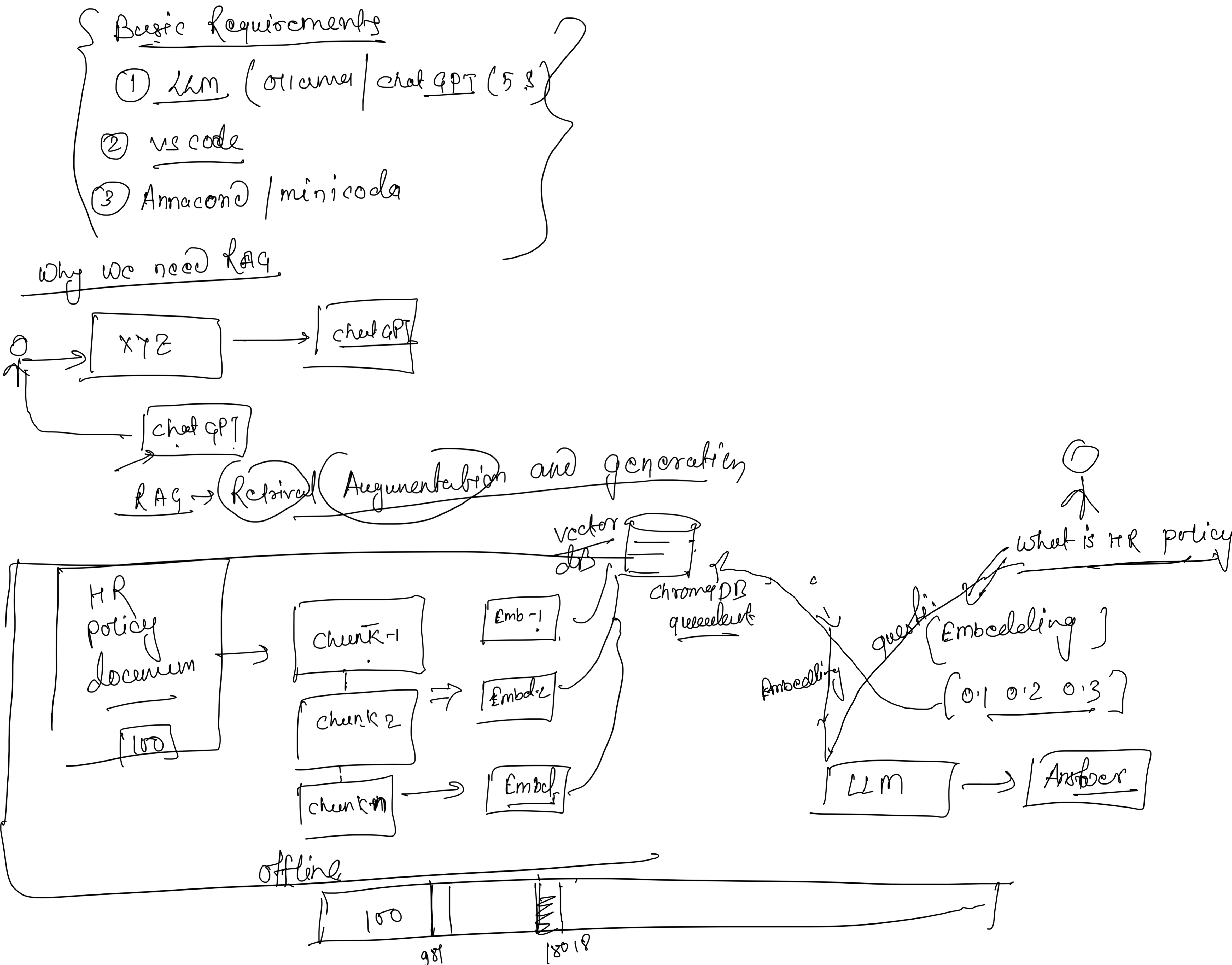
Proposed Syllabus

Day 1: Foundations of RAG
Theory:
Introduction to LLMs and limitations (Knowledge Cutoff, Hallucinations).
What is RAG? Why do we need it?
Core Components: Retrieval, Augmentation, Generation.
Deep Dive: Embeddings (Vector representation of text).
Deep Dive: Chunking (Fixed-size, sliding window).
Practical:
Setting up the Environment (LiteLLM, Ollama).
Basic RAG Pipeline from scratch (No framework magic first).
In-memory Vector Search (using numpy/cosine similarity).
Deliverables:
01_setup_and_basics.ipynb: Setup guide and basic concepts.
02_simple_rag.ipynb: A simple RAG implementation.

Day 3: Optimization & Evaluation (The "RAG of RAG")
Theory:
RAG Evaluation Metrics (Context Precision, Recall, Faithfulness).
Improving RAG:
Query Rewriting / Expansion.
Reranking (Cross-Encoders).
Hybrid Search (Keyword + Vector).
"RAG of RAG" / Hierarchical Indices / Router Query Engines.
Practical:
Implementing a Reranker using sentence-transformers.
Building a Router (deciding between different indices or tools).
Simple Evaluation using an LLM as a Judge.
Deliverables:
04_rag_optimization.ipynb: Reranking and Query Expansion.
05_rag_of_rag.ipynb: Router and Hierarchical approach.

Day 2: Advanced Retrieval & Vector Databases
Theory:
Vector Databases: Why use them? (Chromadb, FAISS, Milvus).
Advanced Chunking Strategies (Semantic, Recursive, Markdown-based).
Retrieval Types: Dense vs Sparse (Keyword) vs Hybrid.
Practical:
Implementing ChromaDB.
Ingesting a larger dataset (e.g., PDF/Documentation).
Experimenting with different chunking sizes.
Deliverables:
03_advanced_chunking_and_vectordb.ipynb: ChromaDB integration and chunking experiments.

Day 4: Multimodal RAG & Production
Theory:
Multimodal Embeddings (CLIP, SigLIP).
Storing and Retrieving Images/Tables.
Vision-Language Models (VLMs) for answer generation.
Practical:
Building a Multimodal RAG pipeline (Text + Images).
User asks question about a chart/image, system retrieves and answers.
Deliverables:
06_multimodal_rag.ipynb: Multimodal implementation.



Day 3: RAG Evaluation, Optimization & RAG of RAG

1. Why Evaluate RAG?

You've built a RAG system, but how do you know if it's good? Evaluation is critical for:

Measuring Performance: Quantify retrieval and generation quality.

Comparing Approaches: Which chunking strategy works best? Which embedding model?

Iteration: Identify weaknesses and improve.

2. RAG Evaluation Metrics

Retrieval Metrics

Context Precision: Are the retrieved chunks actually relevant?

Formula: Relevant Retrieved / Total Retrieved

Example: If you retrieve 5 chunks and only 3 are relevant → 60% precision.

Context Recall: Did you retrieve all the relevant chunks?

Formula: Relevant Retrieved / Total Relevant in Database

Harder to measure (you need ground truth).

MRR (Mean Reciprocal Rank): How high is the first relevant result?

Formula: 1 / Rank of First Relevant Result

Generation Metrics

Faithfulness: Is the answer grounded in the retrieved context?

Check if the LLM is hallucinating or using external knowledge.

Answer Relevance: Does the answer actually address the question?

Answer Correctness: Is the answer factually correct? (Requires ground truth).

Combined Metrics

RAGAS (Retrieval-Augmented Generation Assessment):

A framework that combines multiple metrics.

Uses LLMs to evaluate other LLMs (LLM-as-a-Judge).